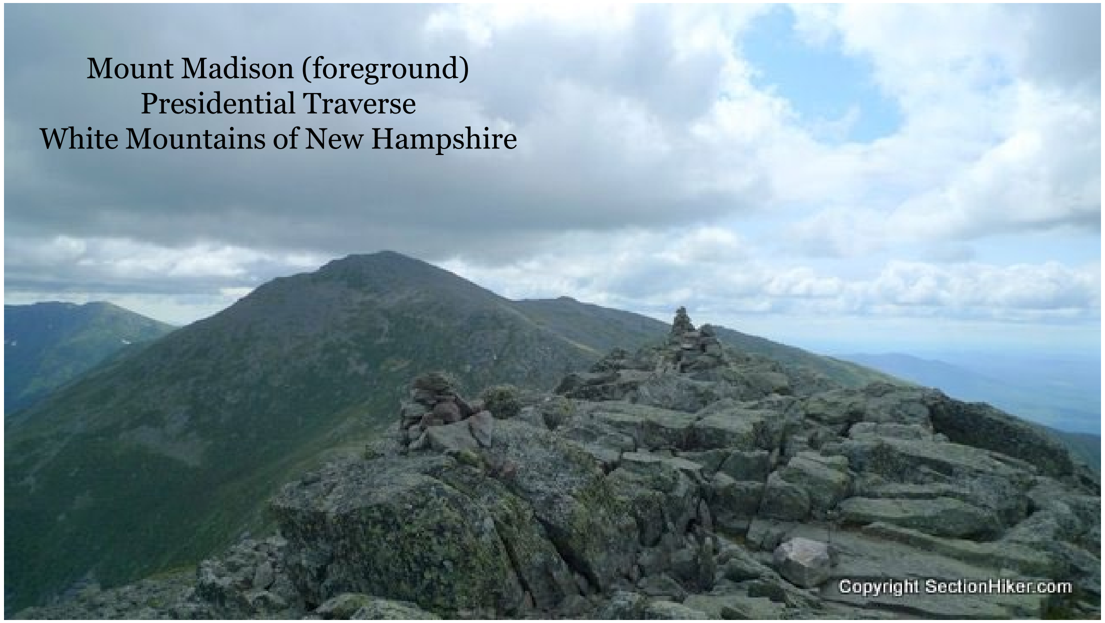
Multivariable Functions

Trim Ch 11, S. 12.1



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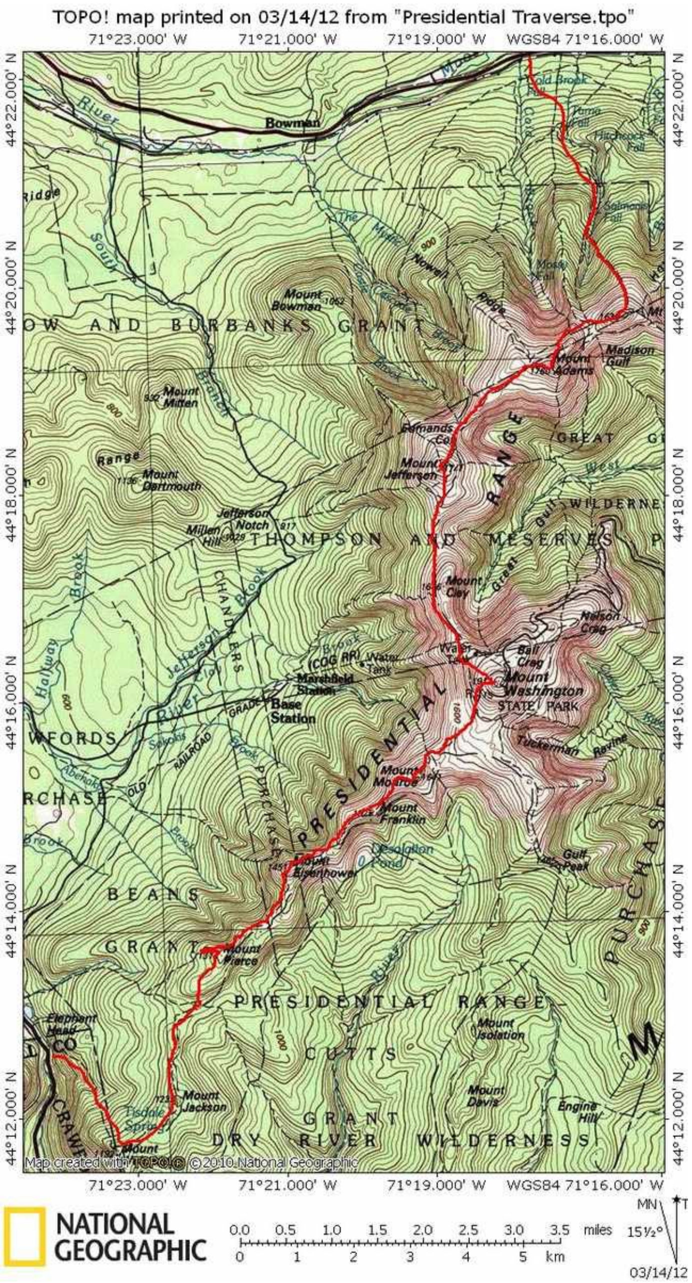


Multivariable Functions

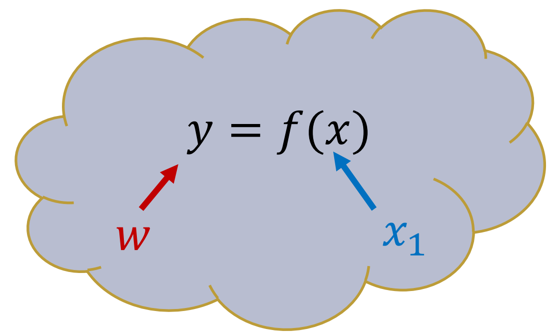
Suppose D is a set of n-tuples of real numbers $(x_1, x_2, x_3 \cdots, x_{n-1}, x_n)$

A **real valued function *f*** on D is a rule that assigns a unique (single) real number "*w*" to each element of D

Function of n-independent variables

$$w = f(x_1, x_2, x_3 \cdots, x_{n-1}, x_n)$$

dependent variable



Domain: D
Range: Set of all w

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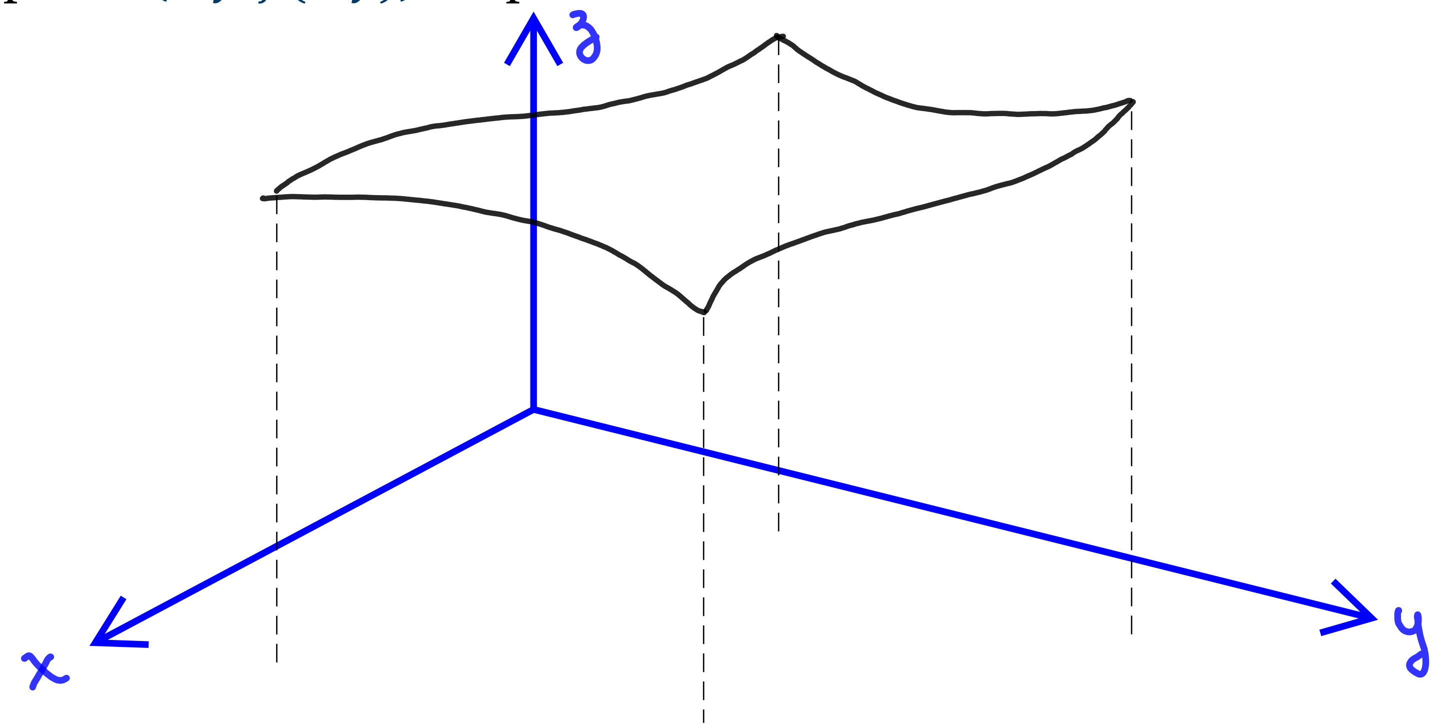
Graph of a Function of 2 Variables

A function of 2 variables $z = f(x, y)$ is the equation of a 3D surface formed by the set of all points $(x, y, f(x, y))$ in space.



1. The value of $z = f(x, y)$ is the height of the surface at $P(x, y)$
2. The domain of $f(x, y)$ is the set of all points in the $x - y$ plane that correspond to a unique value of $z = f(x, y)$

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Contour and Level Curves of Functions of 2 Variables

There are two standard ways to visualize the values of a function $f(x, y)$. One is to sketch the surface $z = f(x, y)$ and another one is to draw and label curves in the domain on which $f(x, y)$ has a constant value.

Example:

Graph $f(x, y) = 100 - x^2 - y^2$ and plot the level and contour curves for $f(x, y) = 0$, $f(x, y) = 51$, $f(x, y) = 75$, in the domain of $f(x, y)$.

Solution:

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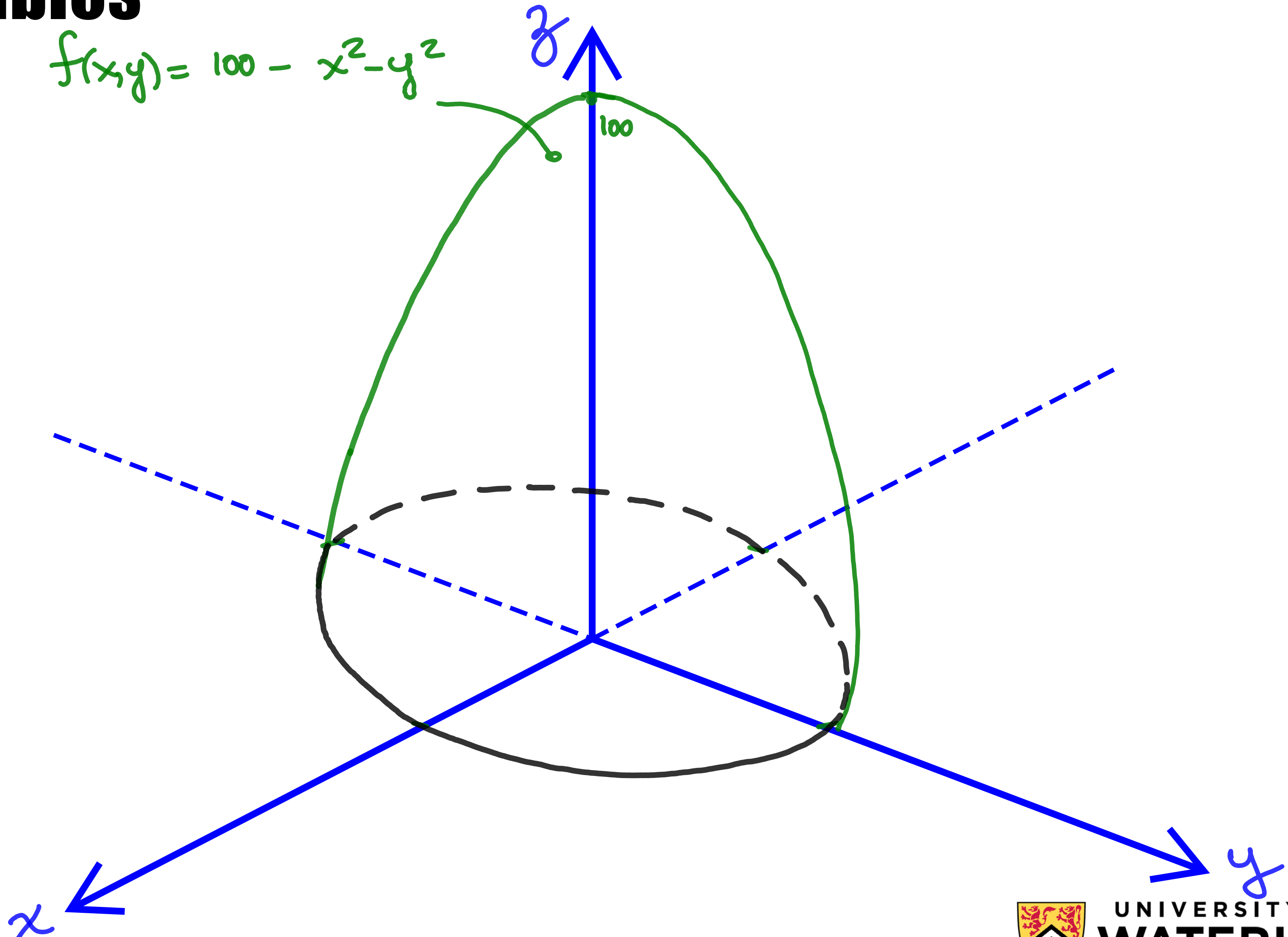
Contour and Level Curves of Functions of 2 Variables



So far …

$z = f(x, y)$ represents the height of a 3D surface at (x, y)

CONTOUR CURVE

A plane $z = c$ parallel to the $x - y$ plane and intersecting a surface $z = f(x, y)$ produces a *Contour Curve*.

LEVEL CURVE

Set of points in the domain where a function $f(x, y)$ takes on a given value c.

CONTOUR MAP

Plots where equidistant level curves are projected in the x-y plane, specifying their z-value

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Graphing Functions of 3 Variables

A function of three variables, f, is a rule that assigns to each ordered triple (x, y, z) in a domain D a unique real number denoted by $f(x, y, z)$.

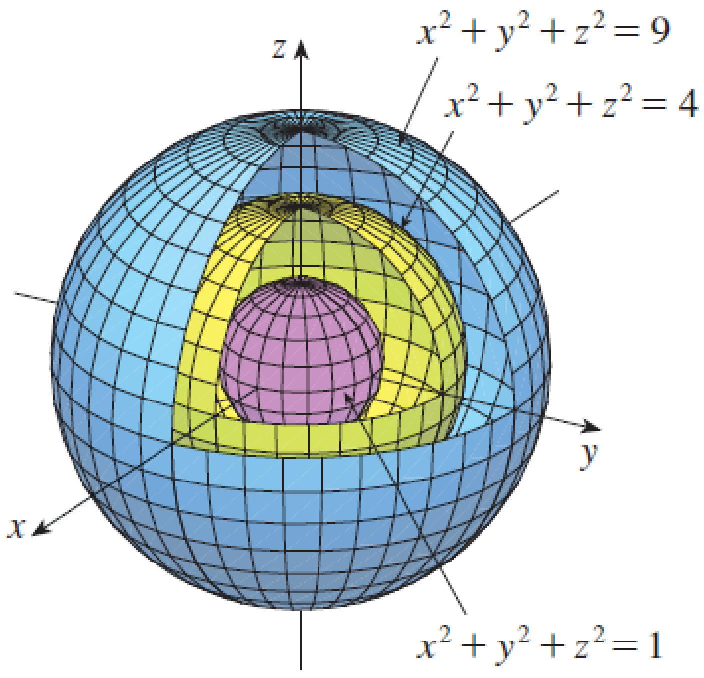
Example:

Graph the function:

$$f(x, y, z) = x^2 + y^2 + z^2$$

Solution:

The level surfaces are $x^2 + y^2 + z^2 = k$, where $k \geq 0$, and they form a family of concentric spheres with radius $\sqrt{k}$



We are not graphing the function here; we are looking at level surfaces in the function's domain.

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Summary

- A **real valued function *f*** on D is a rule that assigns a unique (single) real number "w" to each element of D
- We use multivariable functions to **define real-life relationships**
- **Contour map** is a plot where equidistant level curves are projected in the $x - y$ plane. They are labeled by specifying their constant $z -$ value
- A function of three variables $f(x, y, z)$ lies in the 4D space and it can be "visualized" in 3D space by graphing their **level surfaces**

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Thank you